Computation Signal We observe that linear separability is not a static property of the representation but follows a distinct pulse-like trajectory. As shown in Figure 3ab, we see that the capacity measured by a given node's ground truth answer, namely a "node capacity", spikes when that node is being solved, and then decays back to the baseline. This is in contrast to an alternative possibility that each node's capacity stays up after the node is solved. Essentially, the information about the solution of a node is erased from the residual stream immediately after being solved, to create bandwidth for the subsequent node to be solved. Later, when the node is recalled to solve its parent node, the capacity spikes again, but to a lesser degree Figure 3ab.

Since the LM manages the representations such that only a small number of relevant concepts are separable (the two children node answers, and the current node's answer) at a given time, the dimensionality of the representation can be kept small, assuming that each concept takes different directions according to the Linear Representation Hypothesis. This is reflected by our estimations of both the local (intrinsic) and global dimensionalities, measured by the TwoNN and participation ratios, respectively (Figure 3c). Across the entire CoT, both dimensionalities hover around just 10.

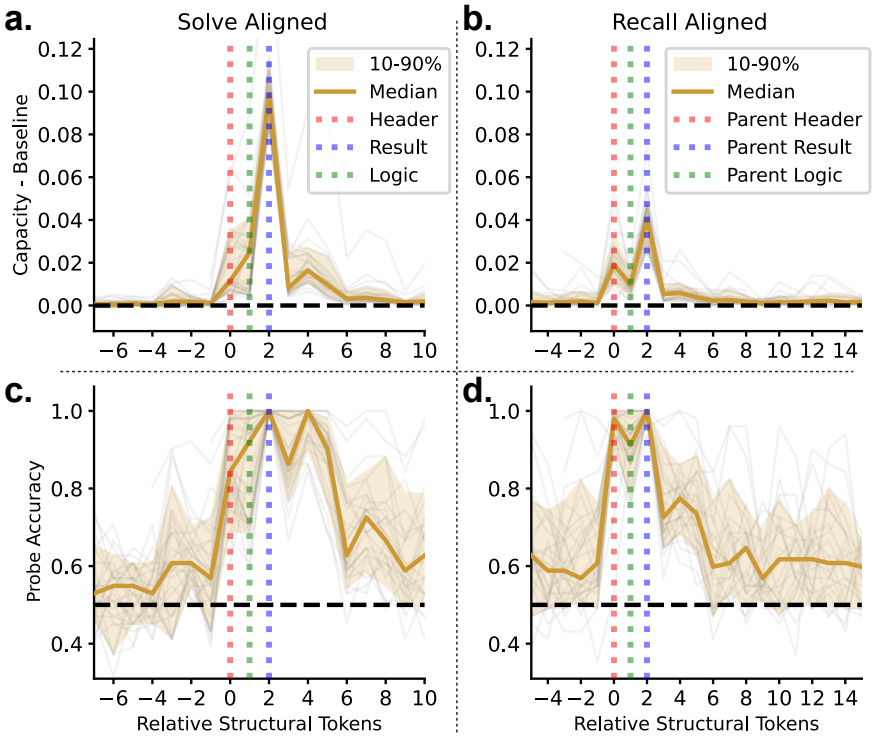

Figure 4: **Dynamics of separability metrics during Solve and Recall events.** Data is aligned to the moment a node is computed (Left column: **a**, **c**) or when it is recalled by its parent (Right column: **b**, **d**). (**a**, **b**) *Manifold Capacity:* Shows a sharp, transient peak at the moment of computation and recall, dropping quickly to baseline in between. (**c**, **d**) *Hard-margin SVM probe test accuracy:* While accuracy also peaks at computation, it exhibits a sustained plateau, decaying much slower than capacity. Vertical dashed lines mark the relative positions of structural tokens (Header, Result, Logic).

To reveal a fine-grained pattern of the canonical trace of the node capacity at the node's solve phase and recall phase, we align the traces at the solve phase structural tokens Figure 4a, and at the recall phase structural tokens Figure 4b. There are three structural tokens, i.e. "header", "logic", and "result", in each phase. We see that the capacity for a target node peaks at the structural tokens immediately *preceding* the generation of the textual answer. The increase in separability begins as soon as the model generates the tokens corresponding to the node's dependencies. This indicates that the linear separability of the concept emerges in the residual stream prior to the explicit token decoding.

Separability Decay and Divergence with Linear Probes Immediately after a node's computation is complete, its Manifold Capacity decays rapidly (Figure 4b). This marks a divergence compared to the accuracy of a Linear Probe and an SVM trained on the same features. While the probe accuracy remains high ($> 90\%$) for the tokens corresponding to the next two nodes being computed, the Manifold Capacity drops near the random baseline. This divergence highlights a difference between the information recoverable by a trained classifier and the information available via linear separation in the current geometry. However, as shown in the Supplementary section, the signal is much noisier and weaker with

the probes, and it can only be revealed when the traces are aligned Figure 4cd (the result for SVM is shown here; the result for Logistic Regression can be found in the Supplementary section).

4.3 Spatiotemporal Localization

We extended our analysis to the full Layer $\times$ Time grid to locate the physical origin of these geometric pulses. Given that the model recreates its residual stream at every token, this allows us to see how the model chooses to expand or not any given piece of information that it might want to do computations on across the layers.

Figure 5 presents the capacity change averaged over all nodes along the reasoning trace, aligned with respect to the moment where the node is solved. We identify two distinct phases:

1. **Reasoning Phase:** At the *header* and *logic* structural tokens of a specific node, the increase in separability originates in the middle layers (Layers 13-15). Then at the *result* structural token (immediately before the answer is spelled out), the capacity strongly increases throughout all layers.
2. **Reorganization Phase:** In the subsequent tokens that pertain to the next node's computation, the evolution is dramatically different. While early layers maintain higher capacity (likely retaining the immediate context), the middle layers (13-15) show a rapid decrease in capacity for the previous node, so that in the later layers (16-32) the information is not readily accessible anymore.

A similar behavior can be seen when looking at the linear probes test accuracy (Figure S.2) for the reasoning phase, however the suppression phase is a lot less marked for them, and the probe accuracy hovers above 90% in later layers for up to two nodes after the computation is done.

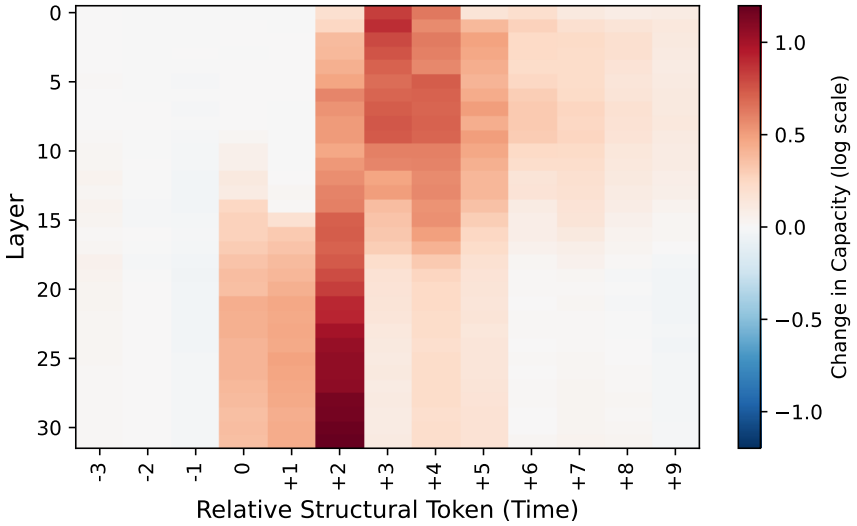

Figure 5: **Spatiotemporal heatmap of manifold capacity change.** The x-axis represents token positions relative to the computation step ($t = 0$). (*During Computation,* $t = 0, 1$) A capacity increase initiates in the middle layers at $t = 0$, culminating in a sharp peak within the deep layers at the token preceding the result ($t = +2$). (*Post-Computation*) For $t > +2$, the early-to-mid layers retain elevated capacity as they retain the immediate context; however, this capacity decreases in the middle layers as the model reorganizes its representation to shift focus to the next node.

4.4 Implicit Compositionality (Silent CoT)

Does the capacity of a node increases when the node is being recalled, simply because it needs to write down that node's truth value (or even the name) during the recall phase, or because of computational need? To check this, we prompted Ministral to silence the textual output of the child node names or values during CoT (see S.I. B). We observe that the child nodes' capacities still spike during the recall phases, strongly indicating that these spikes are for the internal computations, not simply for the textual output.

4.5 Directions of Separation

We saw in Figure 3c that the dimensionality is kept low during CoT, and only a few concepts are separable, minimizing the bandwidth of the residual stream. The question is, what exactly takes up how much space in the residual stream

bandwidth during computation? To test this in our setup, we investigate the directions orthogonal to the max-margin separating hyperplanes.

Figure 3 shows that when a node is being solved, the manifold is separable by the answers of both the left and right child nodes. Interestingly, we find that there are two strongly conserved directions for separating the left and right child node answers, and they are slightly pointing in opposite directions, taking at least two dimensions (Figure S.7-**right**). The fact that the left and right child node directions are strongly conserved indicates the LM truly solves the problem in a compositional manner, applying the same local rule across subproblems. Since the manifold needs to be separated with respect to the parent node as well, the minimum number of dimensions required for the computation is three. In addition, we find that the direction that separates a node's answer while it is being solved is not conserved across nodes (Figure S.7-**left**). Moreover, the directions in which the node is solved and recalled are also not conserved (Figure S.8). See Section C for more details.

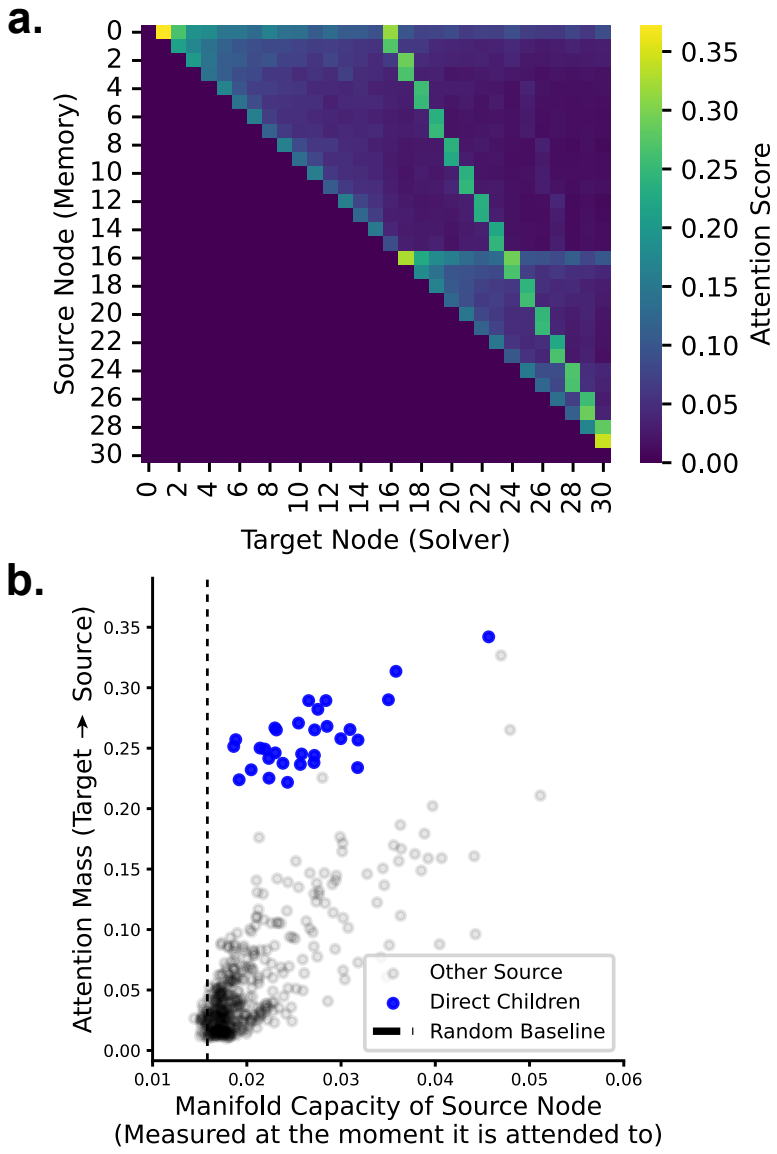

Figure 6: **Correlation between Attention and Manifold Capacity. a.** Attention score heatmap between graph nodes. The x-axis represents the target node currently being solved; the y-axis represents the source nodes in the context. The diagonal structure reflects the model attending to recent context, while off-diagonal points represent the retrieval of specific dependencies. **b.** Scatter plot of Attention Score versus Manifold Capacity of the source node at the moment of attention. We observe a strong positive correlation ($r = 0.723$), specifically for direct children nodes (blue), indicating that the model preferentially attends to previous information when its representation is linearly separable.